Listing 7.11. Visualizing the learned action-value distributions

```
tpred = pred_batch
cs = ['gray','green','red']
num_batch = 2
labels = ['Action {}'.format(i,) for i in range(aspace)]
fig,ax = plt.subplots(nrows=num_batch,ncols=aspace)

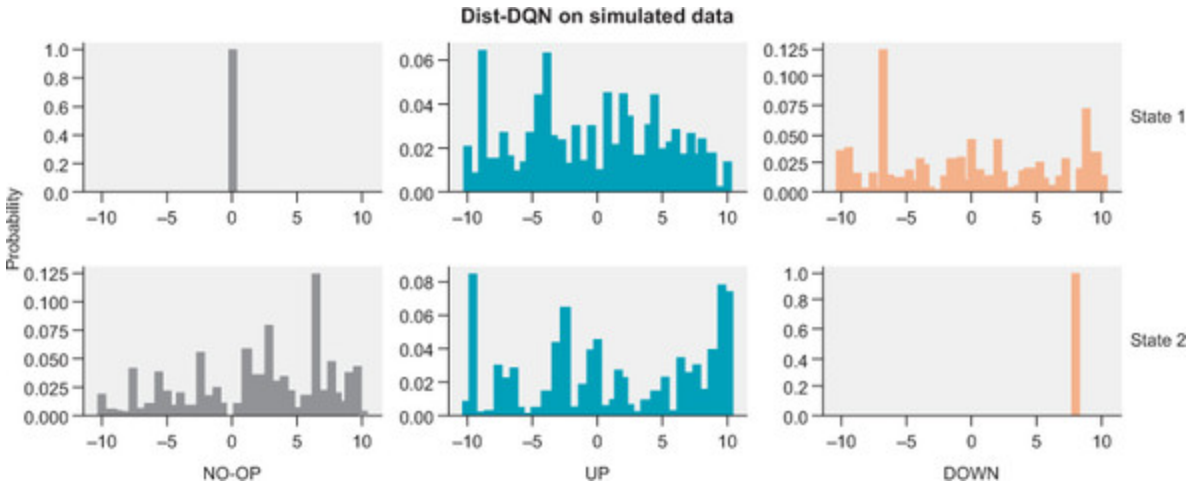
for j in range(num_batch):                    1
    for i in range(tpred.shape[1]):           2

ax[j,i].bar(support.data.numpy(),tpred[j,i,:].data.numpy(),\
                label='Action {}'.format(i),alpha=0.9,color=cs[i])
```

- ***1*** Loops through experiences in batch
- ***2*** Loops through each action

In figure 7.19 you can see that in the first sample, the distribution on the left associated with action 0 has collapsed into a degenerate distribution at 0, just like the simulated data. Yet the other two actions remain fairly uniform with no clear peaks. Similarly, in the second sample in the batch, the action 2 (DOWN) distribution is a degenerate distribution at 10, as the data was also degenerate (a sequence of identical samples), and the other two actions remain fairly uniform.

Figure 7.19. Each row contains the action-value distributions for an individual state, and each column in a row is the distribution for actions 0, 1, and 2 respectively.



This Dist-DQN test has almost everything we will use in a real experiment with Atari Freeway. There are just two functions we need before we get to